

Dazaet Galicia-Badillo

PhD Chemical Sciences
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About me

My research has been focused on the design and synthesis of organic crystals and metal-organic frameworks exhibiting dynamics from the molecular level to the macroscopic level, including rotational motion, photoisomerization, crystal flexibility, salient mechanical responses, and single-crystal-to-single-crystal phase transitions. I bring a strong experimental toolkit: organic synthesis, multinuclear solid-state NMR (^{13}C, ^{19}F, ^{31}P, ^{2}H, ^{15}N), single-crystal and powder X-ray diffraction (SCXRD/PXRD), fluorimetry, and DFT calculations. My long-term goal is to build an independent research program that bridges materials chemistry with applied challenges in energy storage, environmental remediation, or biomedical applications in academia or the private sector.

Honors, Awards & Funding

Nominated* Alfonso Caso Medal nomination - UNAM. 2026
Nominated for recognition of outstanding postgraduate achievement (*outcome pending).

Ph.D. awarded with honors Chemical Sciences - UNAM.
2026

Best Presentation Award - IX Seminario Regional de Materiales Avanzados, UAEH, Mexico, October
2025

Supercomputing resource allocations – National High-Performance Computing Laboratory Search (LANCAD)
2024
Prepared successful proposals securing two allocations of computational resources for research.

Best Poster Award - DynaMIC23, Fribourg, Switzerland
2023
Dynamic Materials, Crystals and Phenomena Conference hosted by Prof. Simon Krause

Doctoral scholarship - Secretariat of Science, Humanities, Technology, and Innovation (SECIHTI) 2022
Support for Ph.D. studies

Master's scholarship - Secretariat of Science, Humanities, Technology, and Innovation (SECIHTI) 2020
Support for Master's studies

Institute of Chemistry Undergraduate Scholarship -UNAM
2019
Awarded for the best undergraduate research project at the Institute of Chemistry

Education

PhD Chemical Sciences - UNAM, Institute of Chemistry 2022 - 2026
Thesis: Molecular and Macroscopic Motion in Organic Crystals and Metal-Organic Frameworks
Advisor: Prof. Dr. Braulio Rodríguez-Molina

M.Sc. in Chemical Sciences - UNAM, Institute of Chemistry 2020 - 2022

B.S. in Chemistry - UNAM, Faculty of Chemistry 2015 - 2020

Technical skills

Experimental: Organic synthesis, including air- and moisture-sensitive reactions (Schlenk techniques); MOF solvothermal synthesis and post-synthetic modification (PSM). Solid-state NMR (Bruker 500 MHz, double-resonance ^{1}H/X MAS probe): ^{13}C CP-MAS, ^{19}F MAS, ^{31}P, variable-temperature ^{2}H and ^{15}N CP-MAS; ^{2}H lineshape simulation; DFT calculation of NMR parameters (CP2K); analysis of ^{13}C T_1 and solid-state ^{1}H and ^{19}F relaxometry data. X-ray diffraction: SCXRD and PXRD, including variable-temperature studies and structure refinement (ShelX). Photophysics: steady-state and time-resolved fluorescence, UV–Vis absorption. Thermal analysis: hot-stage microscopy, DSC, TGA. **Computational & Software:** DFT calculations (geometry optimization, electronic structure, energy profiles, PES) in Gaussian16; periodic DFT in CP2K and molecular dynamics simulations (LAMMPS) on HPC clusters. Crystallography tools: ShelX, CrystalDiffract, CCDC suite. Visualization: CrystalMaker, ChemDraw. NMR processing and simulation: TopSpin, MestreNova, NMR WebLab, Spinach. AI-assisted workflows for research automation and data analysis. **Lab management:**

Purchasing and vendor management; inventory control and lab organization; team leadership and training; funding acquisition (grants/proposals).

PUBLICATIONS (RESEARCH, REVIEW AND PREVIEW)

14. Ernesto Enriquez-Palacios, **Dazaet Galicia-Badillo**, Marcos Flores-Alamo, Jose Luis Belmonte-Vazquez. Imidazole-based Photoswitches by Design and their use for Data Encryption. *ACS Applied Materials & Interfaces*. **2026.** 18, 35, 47895–47906. *[IF: 8.5]*

13. Dazaet Galicia-Badillo, Ernesto A. Hernández-Morales, Yesterday López-Herrera, Erick Hernández-Santiago, Jan Blahut, Rubén A. Toscano, Martin Dracínský, Braulio Rodríguez-Molina. Dissecting Molecular and Macroscopic Dynamics in a Jumping Solvate. *Materials Chemistry Frontiers,* **2026**. 10, 18, 3305-3313. *[IF: 8.1]*

12. Dazaet Galicia-Badillo, Jade Cisneros, Jan Blahut, Alonso Acosta-Vera, Alfonso García-Márquez, Haggeo Desirena, Erika Bartůnková, Ma. Carmen García-González, Ingrid Chávez-Serrano, Joelis Rodríguez-Hernández, Martin Dracínský, Braulio Rodríguez-Molina. Integration of a Molecular Rotor and a Photoswitch in a Metal-Organic Framework. *ACS Materials Letters*, **2026**. 8, 7, 1995-2001. *[IF: 8.7]*

11. Ernesto A. Hernández-Morales, Dayra Barreto-Hernández, **Dazaet Galicia-Badillo**, Rubén A. Toscano, M. Elena García-Aguilera, Braulio Rodríguez-Molina. Crystals as Rockets: Modulation of the Salient Temperature in Cocrystals by Solvent Mixture Composition. *Angewandte Chemie Int. Ed.*, **2026**, e26010. *[IF: 16.9]*

10. Ángel D. Álvarez-Castillo, Alexis Juárez-Morales, **Dazaet Galicia-Badillo**, Mauricio Maldonado-Domínguez, José M. Heredia-Peñaloza, Marcos Flores-Álamo, Jose Luis Belmonte-Vázquez. Rational Modulation of Excited-State Intramolecular Proton Transfer, Aggregation-Induced Emission, and Intramolecular Motion on Polysubstituted Imidazoles (Front Cover). *ChemPhotoChem*, **2026**, 10, e70099. *[IF: 3.0]*

9. Mario Castañón-García, Pedro López-Mendoza, **Dazaet Galicia-Badillo**, Luis D. Miranda. Synthesis of Non-Symmetrical 3,3'-Bicoumarins: Total Synthesis of Arteminorin C, 3,3'-Biisofraxidin and Biscopoletin. *Organic Letters*, **2025**, 27 (21), 5349-5354. *[IF: 5.0]*

8. José L. Belmonte-Vázquez, Durga Prasad Karothu, Carl H. Fleischer, **Dazaet Galicia-Badillo**, Mauricio Maldonado-Domínguez, Robert W. Schurko, Liang Li, Pance Naumov, Braulio Rodríguez-Molina. Low-Temperature Dynamics at Nano- and Macroscales: Organic Crystal That Exhibits Low-Temperature Molecular Motion and the Thermosalient Effect. *Chemistry of Materials*, **2025**, 37, 3373-3383. *[IF: 7.2]*

7. Dazaet Galicia-Badillo, Braulio Rodríguez-Molina. On the Flexibility of 1,4-Bis(phenylethynyl)benzene Butyl Ester Organic Crystal Featuring Mobile Components. *Tetrahedron*, **2024**, 167, 134245. *[IF: 2.2]*

6. Dazaet Galicia-Badillo, José L. Belmonte-Vázquez, Mario Rodríguez, Braulio Rodríguez-Molina, Ma. Carmen García-González. Aggregation-Induced Emission Enhancement and Solid-State Photoswitching of Crystalline Carbazole N-Salicylidene Anilines. *ACS Omega*, **2024**, 9 (36), 38015-38022. *[IF: 4.3]*

5. Alonso Acosta-Vera, **Dazaet Galicia-Badillo**, Armando Navarro-Huerta, Konstantin Stracke, Virginia Gómez-Vidales, Joelis Rodríguez-Hernández, Jack D. Evans, Braulio Rodríguez-Molina. Modulation of the Photophysics and Internal Dynamics in a Zr Metal-Organic Framework by the Inclusion of Fluorinated Guests. *ACS Materials Letters*, **2024**, 6 (9), 4395-4401. *[IF: 8.7]*

4. Dazaet Galicia-Badillo, Paulina Noemí García-Vargas, Braulio Rodríguez-Molina. Preview: The Integration of Multiple Components for Multi-Responsive Dynamic Framework Materials. *Matter*, **2024**, 7 (8), 2766-2768. *[IF: 17.5]*

3. Erick Hernández-Santiago, Jorge Espinosa-Rocha, **Dazaet Galicia-Badillo**, Braulio Rodríguez-Molina. Review: Local and Global Dynamics in Organic Soft Porous Crystals. *Matter*, **2024**, 7 (2), 408-429. *[IF: 17.5]*

2. Manuel Rodríguez-Molina, **Dazaet Galicia-Badillo**, Enoc Cetina-Mancilla, Jorge Cárdenas, Lilian I. Olvera, Rubén A. Toscano, Braulio Rodríguez-Molina, Mikhail G. Zolotukhin. 9-Trifluoromethylxanthenediols: Synthesis and Supramolecular Motifs. *ACS Omega*, **2022**, 7 (16), 13520-13528. *[IF: 4.1]*

1. Aaron Torres-Huerta, **Dazaet Galicia-Badillo**, Andrés Aguilar-Granda, Jacob T. Bryant, Fernando J. Uribe-Romo, Braulio Rodríguez-Molina. Multiple Rotational Rates in a Guest-Loaded, Amphidynamic Zirconia Metal-Organic Framework. *Chemical Science*, **2020**, 11 (42), 11579-11583. *[IF: 9.8]*

TEACHING & MENTORSHIP

Co-director of 4 bachelor's theses (2023-present), covering MOF synthesis, organic crystal dynamics, and luminescent materials. Supervisor of 9 undergraduate short research stays (2022-2025) and 1 high school student.

Scientific conferences & presentations

15 posters (6 international, 9 national) · 9 oral presentations (4 invited lectures, 5 congress talks)

Scientific divulgation

1 article (Boletín IQ-UNAM, 2022) · 2 public lectures (High Schools, 2024–2025) · 1 media consultation (UNAM Global, 2022)

References

Prof. Dr. Braulio Rodríguez Molina - PhD Supervisor**.** Instituto de Química, Universidad Nacional Autónoma de México
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Prof. Dr. Aaron Torres-Huerta - Research collaborator. Universidad Autónoma de Madrid
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